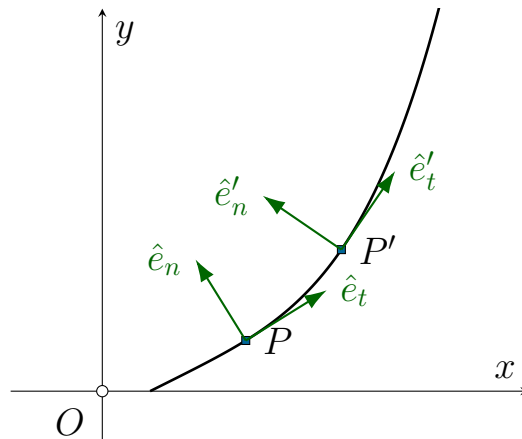


Lesson 4: 2D motion with normal-tangential coordinates

1 The Equations of Motion with Normal-Tangential Coordinates



In normal-tangential coordinates we attach the unit vectors to the particle itself. The t or tangential direction points in the direction of motion at that instant in time. The unit vector in the t direction is denoted as $\hat{e}_t$. By definition its magnitude is one but its direction changes as the particle moves along its path.

The velocity vector can therefore be expressed as:

$$\vec{v}(t) = v(t)\hat{e}_t \quad (1)$$

where $v(t)$ is the speed of the particle at a given time.

We won't derive this, but the acceleration in this coordinate system is expressed as

$$\vec{a} = \frac{dv}{dt}\hat{e}_t + \frac{v^2}{\rho}\hat{e}_n \quad (2)$$

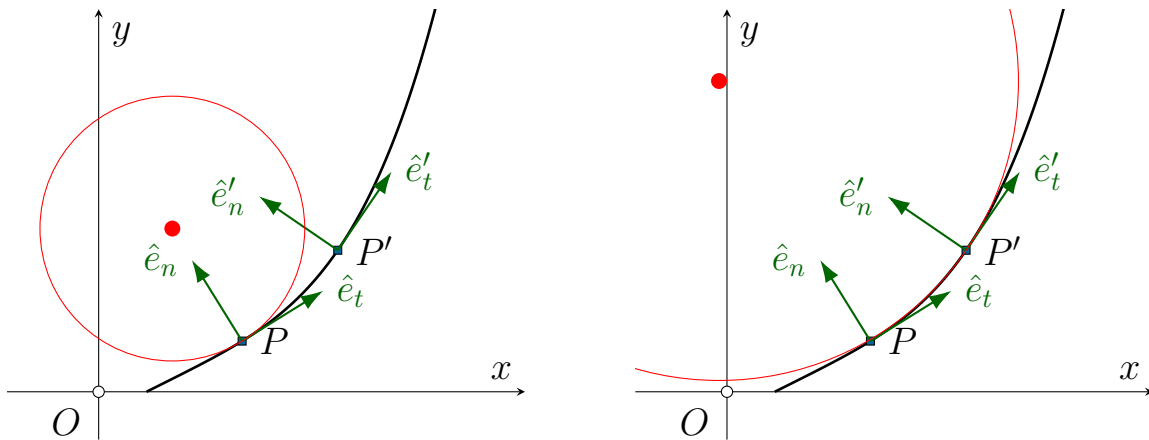
where ρ is the radius of curvature of the path and $\hat{e}_n$ is a unit vector pointing towards the center of curvature.

It will sometimes be convenient to represent the components of the acceleration as

$$a_t = \frac{dv}{dt}, \quad a_n = \frac{v^2}{\rho}$$

therefore

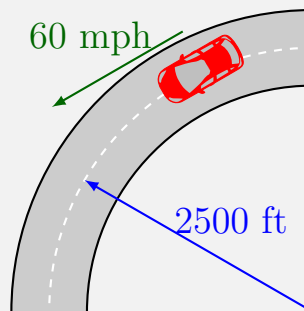
$$\vec{a} = a_t\hat{e}_t + a_n\hat{e}_n \quad (3)$$



The red point indicates the center of curvature and the red circle shows a circle of radius ρ at the point P (left) and P' (right). Notice how $\hat{e}_n$ and $\hat{e}'_n$ point to the center of curvature in each case.

Sample Problem 4.1

A car is traveling on a curved section of highway of radius 2500 ft at the speed of 60 mi/hr. The driver suddenly applies the brakes, causing the automobile to slow down at a constant rate. Knowing that after 8 sec the speed has been reduced to 45 mi/hr, determine the acceleration of the automobile immediately after the brakes are applied.



4.1 – cont

$$a_t = -2.75 \text{ ft/s}^2$$

$$a_n = 3.10 \text{ ft/s}^2$$

$$\vec{a} = 4.14 \text{ ft/s}^2 \quad 48.4^\circ$$

Sample Problem 4.2

A commercial jetliner is traveling a constant 250 m/s when it executes an emergency 180 degree turn. If the turn takes 20 seconds, what is the radius of the curve taken by the plane? What is the acceleration experienced by the passengers?

$$\rho = 1591.5 \text{ m}$$

$$a_t = 0$$

$$a_n = 39.3 \text{ m/s}^2 = 4g$$

Sample Problem 4.1

Assuming a cloverleaf interchange has a radius of curvature of 80 meters at the tightest part of the turn, what is the fastest a car could travel around this curve without experiencing more than $0.5g$ in acceleration? Assume the car is traveling at a constant speed. If the car was instead increasing speed at a rate of 2 m/s^2 , what would be the new overall magnitude of the acceleration experienced by the passengers?

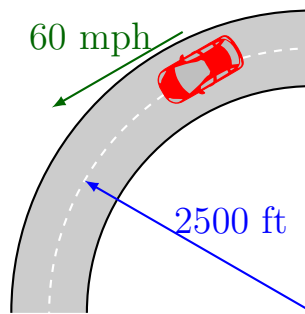


$$v = 19.8 \text{ m/s}$$

$$|\vec{a}| = 5.297 \text{ m/s}^2 = 0.54g$$

Sample Problems

1. A car is traveling on a curved section of highway of radius 2500 ft at the speed of 60 mi/hr. The driver suddenly applies the brakes, causing the automobile to slow down at a constant rate. Knowing that after 8 sec the speed has been reduced to 45 mi/hr, determine the acceleration of the automobile immediately after the brakes are applied.



2. A commercial jetliner is traveling a constant 250 m/s when it executes an emergency 180 degree turn. If the turn takes 20 seconds, what is the radius of the curve taken by the plane? What is the acceleration experienced by the passengers?
3. Assuming a cloverleaf interchange has a radius of curvature of 80 meters at the tightest part of the turn, what is the fastest a car could travel around this curve without experiencing more than $0.5g$ in acceleration? Assume the car is traveling at a constant speed. If the car was instead increasing speed at a rate of 2 m/s^2 , what would be the new overall magnitude of the acceleration experienced by the passengers?

